

Dheeraj Kumar Jallipalli

Software Engineer – AI/ML

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Summary

Software Engineer – AI/ML with 4+ years of experience and expertise in MERN stack development, MuleSoft integration and cloud solutions (AWS). Proven track record in building scalable web applications, enhancing system performance by up to 25% and implementing secure API integrations to drive business growth. Adept at optimizing database management (SQL, NoSQL), leading Agile projects and collaborating with cross-functional teams to deliver high-impact software solutions. Skilled in AWS cloud infrastructure, API development and DevOps practices, with a passion for solving complex technical challenges and improving user experiences. Additionally, experienced in developing and integrating AI/ML models to automate processes, derive insights from data and enhance application intelligence. Proficient in Python, scikit-learn and TensorFlow, with hands-on experience building predictive models and recommendation systems that support data-driven decision-making.

Skills

- **Programming Languages:** Python, Java, C++, C#, JavaScript, SQL, HTML/CSS, Shell Scripting
- **Machine Learning:** Scikit-learn, TensorFlow, PyTorch, XGBoost, LightGBM, Model Monitoring (EvidentlyAI, Amazon SageMaker Model Monitor), Optuna
- **Deep Learning:** CNN, RNN, LSTM, GANs, Transformers, YOLO
- **NLP:** NLTK, spaCy, Hugging Face Transformers, BERT, GPT-4, Entity Extraction, Keyword Tagging, Sentiment Analysis
- **Computer Vision:** OpenCV, YOLOv5, MediaPipe, dlib, ONNX
- **Cloud Platforms:** AWS (SageMaker, EC2, S3, ECS, Fargate, Step Functions, Lambda, Glue), GCP (Vertex AI, BigQuery), Azure ML
- **Big Data & Streaming:** ETL/ELT Pipelines, Apache Spark, Spark Streaming, Hadoop, Kafka, Kinesis
- **MLOps:** MLflow, DVC, Airflow, Docker, Kubernetes, CI/CD, GitHub Actions, Jenkins, Model Retraining Automation
- **Frameworks & Tools:** FastAPI, Flask, Git, Jenkins, JIRA, Postman, ServiceNow, Service First
- **Databases:** PostgreSQL, MongoDB, MySQL, Redis
- **Visualization:** Tableau, Power BI, Matplotlib, Seaborn
- **Model Explainability:** SHAP, LIME
- **APIs & Microservices:** RESTful APIs, Microservices Architecture

Experience

Syneos health, USA

Mar 2024 - Current

Machine Learning Engineer

- Developed a **responsive, high-performance Angular application** using TypeScript, RxJS and Angular CLI, optimized for scalability and low latency on high-traffic platforms.
- Built and deployed a **supervised machine learning model** to classify support ticket texts with **89% accuracy**, utilizing TF-IDF vectorization and logistic regression; containerized and deployed via Flask and Docker.
- Implemented **AI-driven insights and automation** in support workflows by integrating classification models into ticket triaging pipelines, reducing manual intervention.
- Automated provisioning of AWS resources (EC2, S3, RDS) using **Terraform**, reducing manual setup time by **80%** and integrated with **Jenkins** for seamless infrastructure-as-code CI/CD.
- Designed and implemented backend services using **Spring Boot** and **Spring Data JPA**, integrating **PostgreSQL** and leveraging Hibernate ORM for robust data access and performance.
- Developed and deployed **RESTful APIs** using Spring MVC and Spring Boot, enabling smooth integration with third-party systems and ML services.
- Optimized UI performance with **Angular Material** and **Bootstrap**, achieving **30% faster page loads** and enhanced mobile responsiveness with adaptive UI components.
- Automated **CI/CD pipelines** with Jenkins and GitLab CI, minimizing downtime during production deployments and ensuring release consistency.
- Deployed microservices using **Docker containers on Kubernetes**, ensuring **high availability, scalability** and **fault tolerance** across distributed systems.
- Collaborated on integrating **AI/ML-powered components**, such as recommendation engines and anomaly detection modules, enhancing product intelligence and user experience.

Mphasis, India

Jan 2020 - Dec 2022

Software Developer

- Integrated and automated backend and frontend testing using JUnit, Mockito, Jest and Enzyme, ensuring comprehensive test coverage and application stability. Developed a **modern React.js application** using Redux for efficient state management, React Router for dynamic routing and React Hooks to improve component reactivity—resulting in a **35% boost in application performance and user engagement**.
- Designed and deployed **backend microservices using Spring Boot**, leveraging Spring MVC and Spring Security to implement **JWT-based authentication** and **role-based access control (RBAC)**, ensuring end-to-end data security.
- Built and deployed a **machine learning model** to classify customer support tickets into predefined categories with **89% accuracy**, utilizing natural language processing (NLP) techniques. The model was served via a Flask API and containerized using Docker for scalable deployment.
- Integrated the ML classifier into customer service workflows to **automate ticket routing**, reducing manual intervention and improving response time by **40%**.
- Adopted a **microservices architecture**, deploying modular services on **AWS EC2** using Docker containers and **Kubernetes** for orchestration, achieving **high availability and scalability**.
- Enforced strong **OAuth 2.0 security protocols** using Spring Security to protect user data and ensure secure authentication and authorization flows.
- Collaborated with cross-functional teams to prototype and integrate **AI/ML-driven features** such as intelligent search, ticket prioritization and anomaly detection using Python (scikit-learn, pandas, NumPy), improving product intelligence and decision-making.

Education

Masters in Artificial Intelligence

Jan 2023 - Jan 2024

Saint Louis University, Saint Louis, MO

Bachelors in computer science and engineering

Jun 2017 - Mar 2021

Vellore Institute of Technology, Amaravati, AP, India